

Institute of Actuaries of India

Subject SA6 – Investment

October 2015 Examinations

INDICATIVE SOLUTIONS

Solution 1:**i)**

The risk model will need to be detailed to take into account all material risk factors arising from the insurance company's investment portfolio.

There should be coverage of all asset classes within current assets and likely future assets.

Additionally, the risk model is likely to be used to measure investment risks arising from the insurer's liabilities.

The model will need to realistically model the dependencies between the dominant asset classes, including any tail dependency.

The model needs to be able to be run within a few hours, which entails compromises between model complexity and available data, and is subject to limitations on computing power.

The model needs to have reliable data feeds from the investment management systems and other data sources to ensure that it can be run with minimal data preparation and manual intervention.

The important information to be published for management report could include:

Liability and asset position

The duration of assets vis a vis duration of liabilities split into various lines of business

The sensitivity of the asset vs liability position with respect to important parameters like interest rates, performance of equities, movement in guaranteed liabilities etc.

The cost of hedging (if company hedges its portfolio) vs the benefits of doing so.

The results of scenario testing giving the movement of related parameters at a time.

[5]**ii)**

There are various reasons an insurance company's model may differ from other companies.

Since the insurance company has long term liabilities, the investment is done with an aim to meet the long term nature of liability. For the purpose of this the life insurance will have normally aim of having long term strategic asset allocation.

Such a strategic asset allocation will only be reviewed infrequently, rather than daily, so it is possible for a process that has significant run-time or requires manual intervention to be followed.

Another unique aspect of life insurance company would be the reinvestment risk on future premiums which would be received upto 20-25 years. General Insurance companies would not have renewal premiums. Banks have a small proportion of liability in the form of recurring deposits wherein they give guaranteed value on maturity on the future deposits too but the tenure generally does not exceed 5 years.

The asset allocation will need to consider multiple portfolios, and may involve some form of optimisation.

The asset liability analysis is likely to consider multiple points on the expected returns rather than just the losses that could occur under the stressed scenarios.

Sensitivity analysis of assumptions (demographic and financial) may be carried out.

The ALM model may include a more complex dependency structure to capture the different levels of dependency / diversification at different points on the return distribution.

[5]

iii) a)

The key objectives that investment policy should aim to achieve are given below:

- Ability to meet the future liability as and when they fall due
- Avoid year on year volatility in provisioning
- Maintain future solvency levels
- Value creation with optimum level of cost
- The level of risk (performance mismatch between assets and liabilities) that the company is prepared to take.

[2]

iii) b)

For conducting ALM testing

Suitable assumptions to be used in the study need to be agreed.

Data on the liabilities needs to be collected to carry out the projections. For detailed liability analysis data on individual policies is required to build up an accurate assessment of the future cashflow projections.

The overall nature of the liabilities is considered — an analysis of current provisioning, maturity and cash flow projections under different scenarios is considered.

An analysis would be carried out to identify how the fund might progress in the future if different investment strategies were adopted.

Different asset mixes would then be analysed in more detail to assess the risks (relative to the liabilities) and the rewards of each alternative under consideration.

The results would be summarised and presented.

Compared to a traditional actuarial valuation, an ALM provides much more information in various dimensions:

- providing projections into the future (*time* dimension)
- providing some estimate of the range of likely outcomes (*probabilistic* dimension)
- indicating the effect of changing investment strategy (*asset mix* dimension)

[3]

iv)

The suitability of any investment will depend on the liability profile of the Company.

If the company has been writing guaranteed long term business then it is justifiable to have majority of the investment in fixed income securities and some investments may be held in the money market for liquidity purposes.

However for participating business, if the guarantees are lower the company would like to invest in equity to provide better returns to the policyholders in the long term. The equity as an asset class may provide some diversification too and the portfolio may give better risk adjusted returns in the long term.

Hence the life insurance company may consider taking risk on equities in lieu of extra return it can generate. The appetite of taking risk will depend upon:

- i) The level of guarantees offered in the product
- ii) The level of terminal bonus allowed in the product, higher the terminal bonus greater will be the flexibility to invest in the equity
- iii) The expected return on equity
- iv) The overall level of surplus held in the funds (which may give investment freedom to fund manager to make extra returns for the shareholder in the non-participating segment as well)

[5]

v)

The most basic aim of investment is to achieve an optimal trade-off of risk and return by allocation of the portfolio to appropriately diversified combinations of assets.

The precondition for such an optimal trade-off is ability to attain the efficient portfolios, where there is no possibility of increasing return without increasing risk, or of reducing risk without reducing return.

For this purpose we will project equity returns in various scenarios by changing the percentage allocation to equities. The returns of the scenario will then be compared to liability profile of the business. The correlation of equity with other asset classes needs to be considered and returns of different asset classes will be projected.

It may be seen that due to inherent volatility in equity class the increased returns will come at a cost of increased risk.

Thus an optimum level of equity can be reached by projecting the balance sheet in various scenarios including different product mix and investment mix.

[3]**vi)**

An optimum level of equity may prove a better match to the liability specially the increasing level of expenses. Especially for participating business the company will be able to support higher level of bonuses by making extra returns on the fund.

The impact will be quantified in terms of risk-return trade-offs and can be evaluated with and without constraints—for example, duration mismatch tolerances, exposure limits, and so on—to better understand their impact.

However the asset liability position of company will largely depend upon the level of surplus held under various investment scenarios and its effect on regulatory and economic capital.

[2]**vii)***Futures*

Futures contracts are agreements between two parties to trade an asset at a certain future time for a certain price.

Futures are normally traded on an exchange.

The exchange specifies certain standard features of the contract (eg the delivery date – when the contract is settled) and provides a mechanism whereby both parties have a

guarantee that the contract will be honoured.

Consequently there is less credit risk as exchanges are generally AAA rated.

Examples:

equity index futures

bond futures (Gilt futures, Bond Futures).

Generally cheaper and easier to trade than forwards due to greater liquidity in the futures markets and consequently lower spreads.

To hedge the risk in the portfolio a short futures position is required

Forwards

An agreement to buy (or sell) an asset at a certain future time for a certain price.

Generally traded OTC (over-the-counter).

Examples of forwards

currency forwards

FRAs (forward rate agreements) [2 x ½ for two examples]

In this case to hedge the risk, the forward agreements would be to sell a forward on a Long-dated bond (either government or corporate).

Swaps

A *swap* is an agreement between two parties to exchange cash flows in the future.

The agreement defines the dates when the cashflows are to be paid and the way that they are to be calculated.

Usually, the calculation of the cashflows involves the future values of one or more market variables.

Examples

Interest rate swaps

Currency swaps

In this case, an interest rate swap would be appropriate. The life insurance agrees to pay the counterparty cashflows equal to interest at a floating rate on a notional principle for a number of years. At the same time, the counterparty agrees to pay the scheme cashflow equal to a predetermined fixed rate on the same notional principal for the same period of time.

The currencies of the two sets of cash flows are the same. (This is typically referred to as a “plain vanilla” interest rate swap).

The notional principal is used only for the calculation of interest payments. The principal itself is not exchanged.

Typically traded OTC.

Options

Options are traded on exchanges and the over the counter market.

Examples

Call Option: gives the holder the right to buy the underlying asset by a certain date for a certain price.

Put Option: gives the holder the right to sell the underlying asset by a certain date for a certain price.

The agreed price in the contract is known as the exercise price or strike price and the date is known as the expiration date or maturity date.

In this case buying call options on the bond market would be required to hedge. If interest rates fall, and liability values rise more quickly than assets, then bond prices will rise. The call option will become more valuable.

However, call options may also have an advantage in that they still leave the portfolio open to upside potential if interest rates rise. The scheme would benefit from its shorter asset duration.

[10]

viii) (a,b)

FPO is Follow – up Public Offering. It is issuing of shares to investors by a public company that is already listed on an exchange. An FPO is essentially a stock issue of supplementary shares made by a company that is already publicly listed and has gone through the IPO process.

FPOs are popular methods for companies to raise additional equity capital in the capital markets through a stock issue.

IPO is Initial Public Offering and FPO is Follow-up Public Offering.

A company makes an IPO for raising money for the first time and an FPO for adding to the capital already raised through initial public offerings.

Initial Public Offering is the first sale whereas the Follow-up Public Offering is the second sale for expanding businesses.

IPOs are risky investments as an individual investor cannot predict what will happen to the initial trading in the coming days.

In the case of FPOs, the risk is lower as an investor already has an idea about the investment and future growth of the company.

However investments made into Initial Public Offerings are more profitable than Follow up Public Offerings.

A company brings out an FPO for further growth. If a company is coming out with an FPO, it also means that the company is short of funds. FPO is raised for more funds or money or for establishing new projects.

FPO vs OFS

The OFS route is used by promoters to sell their shares to the public. The cash raised goes straight to the promoters. An OFS can be done only by the top companies by market capitalisation.

On other other hand any listed company which wants to raise fresh funds or whose promoters want to sell stake can use the FPO route.

An OFS is open only for a single trading session. The company defines a floor price (though declaring this is not mandatory) below which bids will be rejected. It's up to investors to decide what price they want to buy the share. An FPO, on the other hand, defines a price band within which bids should be placed. The offer will be open for at least three and up to 10 days.

In an OFS, one will know the allotment a couple of hours after the offer closes and settlement is done the day after this. OFS also allows you to place multiple bids at different price points. Allotment in an FPO can begin only after the issue closes. The time taken for allotment to be made and the new shares listed is usually up to 12 days. Multiple bids aren't allowed.

Since OFS closes in a day, one needs to pay the entire money upfront when placing the bid. Applications in an FPO will not see the whole amount going out of your account unless you receive allotment.

The company would be interested in going the FPO route to increase the participation from retail investors as the FPO route provides an advantage on factors like pricing and allocation.

OFS issues only have a floor price, while FPOs give a price band within which the investors could bid. The latter offers them more certainty on pricing. The FPO route requires a fixed minimum allocation to retail investors; that is, every investor will receive a certain minimum allocation of shares, if available. OFS issues do not have any such certainty.

[3+5=8]

[43 Marks]

Solution 2:

i) Pensions market in India

a) Opportunities

- Demographic Transition - Full mid-section with predominance of young and middle age and significant volume at the older age. Structure is in rather rapid transition to a more aged population with more than 30 per cent of older persons by 2050.
- Youngest Population - India has the world's highest number of 10 to 24-year-olds. Rapid expected increase in wealth and potential for huge market for pension solutions with both an accumulation phase and eventual income phase.
- Fertility Rate - Fertility rate of India to decrease further from current level of 3 as experienced by developed nation. This further going add to the increased need of financial security at old age.
- Life Expectancy - Current level of life expectancy of India is low. India to reach a level as observed in a developed nation by 2045-50.
- Dependency - more nuclear family structure indicating requirement for support at older age. Old age dependency ratio of India to reach a level observed by a developed nation by 2050.
- Per capita Health Expense - Per capita health expenditure tends to increase with ageing population and old-age dependency ratio.
- Regulation - Annuity purchase mandate post accumulation phase. Better regulations for insurance & pension providers like FDI.
- Others – Govt-sponsored Pension schemes like NPS, Atal Pension Yojna. Media as an alternate sales channel.

[3]

b) Regulatory challenges

- Life insurer authorized to provide annuities, restricting the number of players.
- Indian Assured Lives Mortality (2006-08) table used from April, 2013 to compute annuity which may not be appropriate for the target population.

- Longevity and mortality risk are not allowed to offset for capital computation in current statutory regime.
- Insurers in India allowed to hedge interest rate risk with the benefit limited to 12 months under the rule of forecasted transactions.
- Contribution towards pension scheme covered under 80C which also includes investment in ELSS, life insurance premium, home loan principal repayment, ULIPS, fixed deposits.
- Income from annuity treated as income and fully taxable.
- Service tax applicable on the initial annuity investment, currently 3.5%.
- Discount and loading should not exceed 30% of the approved premiums.
- Approval required to change the annuity rate by more than 10% increasing the asset liability mismatch during large change in interest rate.

[3]

c) Other Risks & Challenges

- Longevity risk - Difficult to predict life expectancy based on past experience as there are medical advances. Lack of credible data.
- Market risk - Annuity rate depends on the market condition at the time of buying resulting in lower amount of benefit during bearish phase.
- Inflation Risk - Inflation may erode the real benefit of regular income from annuity.
- Liquidity Risk - Most of the investment in pension scheme is not accessible until retirement. Even on retirement only part of the accumulated amount can be opted as cash lump sum.
- Operational risk - Model and parameter risk in predicting the long term interest rate, life expectancy and expenses related to the scheme. Higher chance of mismanagement of fund over the longer period of the policy.
- ALM - longest tenor of government bonds available is 30 years. Inflation index bonds with maximum tenor of 10 years. Resulting in high rollover risk. Longevity swap market not mature. Longevity index used may not be exact replica of underlying population resulting in basis risk.
- Alternative investments are available that provide better return than low yielding annuities like equity release schemes and self-managed trusts which are more popular than annuities.
- Traditional investment of relying on property as a financial security in old age.
- Conflict of interest between pension provider and policyholder (unlike life insurance provider and policy holder)
- Agents are the preferred sales channel currently. Present incentive structure for agents is often skewed in favour of non-annuity insurance products.

[3]

ii) Comparison of different schemes.

- a) Final salary DB schemes with no employee contributions (or fixed employee contribution rate)

- A defined benefit final salary scheme is the most risky type of retirement benefit arrangement for an employer.
- Actuarial valuation is difficult as it includes allowance for future salary increases even though no such increases have yet been earned.
- However, when management decisions are taken about employee salary increases, little regard is made to the knock-on effect on the cost of pension benefits.
- In principle, a procedure could be established to detect the cost implications at the time of making decisions about employees' salaries. For example: Install a management discipline that recognizes the past service cost of awarding salary increases when they are given.
- Gratuity schemes in India are DB schemes

[2]

b) Pure DC scheme

- The cost is within the employer's control and all investment and annuity risk is with the employee.
- A DC that invests in assets that provide a secure return (as is the case in India) is essentially the same as a cash balance scheme.
- The accumulation of the "cash balance" is determined according to the yield earned on the scheme's investments.
- Indian income tax rules require retirement benefits from defined contribution pension schemes to be taken as annuities, except to the extent of one-third commutation for cash at retirement.
- Provident Fund contribution is in the nature of DC scheme.

[2]

c) Hybrid of both DB and DC - this would involve a final salary benefit with money purchase underpin

- Increase in administration costs.
- The complexity leads to lack of understanding on the part of employees, which in turn reduces the extent to which employees value the cost to the employer of providing the benefits.
- The efficiency of operating such a complex mixture of schemes is questionable, since the mixture is determined largely by tax considerations rather than a holistic approach to providing for employees' security in retirement.

[2]

iii) Longevity risk

The liabilities of defined benefit pension plans are calculated using assumptions, including an assumption about the life expectancy of pension plan members. Assumptions related to life

expectancy are generally based on industry mortality tables. The risk that pension plan members will live longer than predicted is referred to as longevity risk.

To deal with longevity risk:

- Carry out an investigation into company's own longevity experience and adapt the mortality table appropriately. This would reduce the element of longevity risk resulting from any potential understatement of current liabilities. The risk remains, however, that even revised longevity assumptions could be incorrect over the longer term.
- Enter into a longevity risk hedging contract like a longevity swap. This will provide the pension plan with a risk mitigation strategy that is focused only on longevity risk while retaining interest rate and investment risk.

[3]

iv) Risk for the annuity provider if it were to enter into a longevity swap mitigation plan

- Counterparty Risk - the risk that the counterparty will not live up to its contractual obligations. This can be mitigated by:
 - Payment flows between the counterparties should be netted periodically.
 - Collateral can be used to further mitigate counterparty risk.
 - Ensure that the pledging of assets is permitted by the terms of the contract/trust agreement.
 - Access the regulatory regime to which the counterparty is subject, including whether assets are required to be held by the counterparty in respect of the contract and any applicable capital requirements.
 - Consider the strength of the counterparty before entering into a longevity swap.
- Roll-over Risk – Longevity swaps that are entered into for a shorter time period than the liabilities covered introduce the element of rollover risk to the pension plan. Specifically, although the contract protects the plan against changes in mortality over the period of the contract, it does not cover the risk of having to enter into a new contract once the contract expires. Rollover risk is the risk that entering into this new contract will be more expensive, as may result from changes in mortality expectations. This can be mitigated by:
 - Structuring the contract for the remaining life of the covered populations
 - Structuring the contract for a fixed period that is long enough to cover the life expectancy of the majority of the covered population.
- Data/ Model/ Assumptions Risk – the risk that the population, model and assumptions used to price the 'fixed' leg of the swap are incorrect and you end up with an 'out-of-money' swap. This can be mitigated by:
 - Including provisions in contract to have 'fixed' leg revised if there are any issues with the pricing assumptions. This need not be very frequent (once in a year or 2 years.)

- Legal Risk – the risk that contractual obligations are not understood correctly and hence not met. These are custom-made contracts and not traded on an exchange. Therefore exit costs can be very high. This can be mitigated by:
 - The parties negotiating the contract must pay close attention to the terms of the contract.
 - Plan administrators should fully understand the terms and risks of the transaction.
 - Plan administrators should seek legal advice before entering into a longevity swap.

[4]

v) Considerations that the plan administrator and trustees need to make (any five is ok)

- Cost - Understand how much risk is being hedged versus the cost of hedging that risk. Before entering into the longevity swap, conduct an investigation into the pension plan's own longevity experience which would help understand the amount of risk that is to be hedged and would help ensure that the risk reduction justifies the cost of entering into the contract. Ensure that you receive market-based prices from a number of potential counterparties to compare, thus helping to ensure that a fair market price is paid.
- Acceptability – the longevity swap should be in the best interests of beneficiaries and in accordance with the pension plan contract and the regulations.
- Administrative Complexity – this could include processes for ensuring that the payments between the pension plan and counterparty are timely and correct, processes for the posting of collateral, the potential need for periodic monitoring of the financial strength of the counterparty, etc.
- Duration – Short duration of the swap will expose the pension plan to unhedged longevity risk when the contract expires. A very long-term contract may force the pension plan to lock itself into payments to the counterparty for a long period of time, which could limit flexibility to respond to changing circumstances in the future. Issues related to the duration of a longevity swap could potentially be mitigated through appropriate terms in the contract.
- Liquidity - The plan administrator should consider the fact that it will be difficult to sell or cancel a longevity swap once it has been entered into. Unlike more commonly used swaps, such as interest rate swaps, there is currently no active, liquid market where longevity swaps are traded. In evaluating the risks and obligations associated with a swap, the plan administrator should consider the terms under which the contract may be modified or terminated.
- Actuarial Valuation Implications - include margins for adverse deviations (counterparty, expense, inflation) so that the swap is priced appropriately.
- PFRDA's considerations – Should comply with all regulations. Should be permissible under plan's Statement of Investment Principles. There is no specific requirement to inform plan beneficiaries of the existence of the contract. Plan administrators continue

to be responsible for paying beneficiaries' benefits, including in the event that a counterparty fails to make the agreed upon contractual payments to the pension plan.

[5]

vi) Comparison of the two alternative approaches with recommendation

a) Permanently hedging:

- Removes the risk of not having a hedge in place when you need it.
- Usually will hedge when don't need to, i.e. at time when cost of hedging is lower (i.e. lower volatilities)
- Because you are always hedged even when you don't need to be is quite expensive. It provides a permanent drag on performance of equity portion of the portfolio.
- Will always be in place over financial reporting (interim and year-end) periods, which will assist in amount of capital need to hold

b) Ad hoc

- This could form part of the tactical asset allocation strategy.
- May be forced to hedge – this will typically be when hedging is quite expensive due to high volatility in the market
- But hopefully this will only happen once every 2 to 3 years which means that over time it will still work out cheaper than option above.
- However, given human condition of fear and greed may hedge at exactly the wrong time, e.g. not hedging when you should in the hope that the equity market will improve and then being forced to hedge at the bottom of the market
- Both when volatilities are high and when the level of the market is so depressed that there is very little downside risk to the market
- Therefore important that use a rule based approach, within a risk management framework, where equity exposure will be hedged at a pre-specified level.
- This level should not be too low given the fact that markets can move quite quickly and it may take time to implement the hedges.

c) Recommendation

- Ad hoc approach is better
- Provided have risk management framework in place whereby will hedge at specified funding level
- The sacrifice in returns is less; it is unreasonable to expect that a permanently hedged portfolio will produce returns comparable to unhedged equities; the return is more likely to resemble that of bonds.

[6]

vii) Relative merits of using short-dated (three-month) rather than long dated (nine-month) at-the-money options.

- European options only payable at maturity.

- Nine month options protect for a longer period of time
- And are therefore more expensive than three month options (this could typically be 5.2% vs. 3.7% under normal circumstances)
- However, three-month options will be more effective hedges against market dips than nine-month options, since they are more sensitive to market movements.
- E.g. a 10% drop in the equity market on day one could result in the three-month put option returning 4.1%, while the nine-month put option might return 3.4%
- Will depend on what one is trying to achieve. If are you trying to protect funding levels at a specific date: e.g. at year-end (because of CAR requirements). If so most effective hedge would be one maturing on or just after year-end (because optionality is close to one, assuming it ends in the money)
- Shorter dated options better are more sensitive to falls in market, whereas with nine-month options market may have recovered at maturity and less sensitive to short term movements in market (in case want to cash option in)
- There is generally more liquidity in short-term options. But they would have to be rolled more frequently at prices that are uncertain. Longer dated options lock in the price now.

[4]

viii) Use of single stock options rather than index options to hedge the equity portfolio.

- Objective is to minimize tracking error relative to actual portfolio and to increase effectiveness of hedge.
- Most managers manage to NIFTY or SENSEX, so possibly best to hedge the benchmark and manage the risk of deviation from it.
- But need to analyze actual equity portfolio especially if it differs from the benchmark significantly. Index hedges would not cover exposure to foreign markets, for example.
- A combination of indices might mimic the portfolio better.
- Also take into account liquidity of hedging instrument, NIFTY or SENSEX options are highly liquid.
- Single stock options are administratively complex. And they are more expensive because volatility of individual stocks is far greater than that of indices or portfolios.
- Difficult to obtain options on the smaller stocks. Only hedging the largest stocks can introduce big tracking error.
- Single-stock option could play a role if the manager has a strong view on a stock and want to add value to the portfolio by betting on that view but this doubles up the risk of the manager's view being wrong and negates the intention of being hedged against unforeseeable risks.

[4]

[41 Marks]

Solution 3:

i) Contractionary monetary policy is when the Central bank uses the rate and its other tools to slow economic growth. Contractionary policies are enacted by a government to reduce the money supply and ultimately the spending in a country.

Therefore, the Central bank would only implement contractionary policy to fulfil its primary mandate of preventing inflation.

Increasing Repo rate:

Increasing the interest rate at which the Central bank lends will also increase the rates at which banks lend.

This will increase the rate that banks charge each other to borrow funds to meet the cash reserve requirement.

This requirement is the amount the cash reserve requires banks have on deposit each night when they close their books

Raising the interest rate is contractionary because it decreases the money supply. That's because banks charge higher interest rates on their loans to compensate for the higher interest rate. Businesses borrow less, don't expand as much, and hire fewer workers, decreasing demand. They are also less likely to raise prices, putting an end to inflation.

Increase in reserve requirement:

The Central bank could raise the reserve requirement, which is also contractionary monetary policy.

Banks are required to keep a reserve of cash to meet withdrawal demands. If the reserve requirements are increased, there is less money for banks to lend out. Thus there is a lower money supply.

Open market operations:

The third tool the Central Bank uses in contractionary monetary policy is open market operations. That's when the Central bank buys or sells its holdings of government securities. To implement contractionary monetary policy, the Central bank sells securities to one of its member banks. The bank pays for the securities by adding to its reserves. This reduces the money it has available to lend, so it will charge a higher interest rate.

Hence sales of government securities shrink the funds available to lend and is likely to push the interest rates up.

[4]

ii)

a) Indian Currency:

Any rise in interest rate has a direct impact on the country's currency. Rise of interest rate in the USA will strengthen the dollar. A strong dollar will attract money from other markets causing a ripple effect.

Hence a strengthening currency may have a significant impact on Indian financial system.

The extent of this fall will depend upon strength of foreign currency reserves held and the growth prospects of Indian Economy. Given that the said fundamentals are very strong for India, the depreciation in rupee will not be as big as those for other emerging markets.

Funds may flow out of both debt and equity markets, and put under pressure the rupee.

[3]

b) Stock Market:

When the Fed increases the federal funds rate, it does not have an immediate impact on the stock market.

Instead, the increased federal funds rate has a single direct effect - it becomes more expensive for banks to borrow money from the Fed.

Increases in the federal funds rate also cause a ripple effect, however, and factors that influence both individuals and businesses are affected

The first indirect effect of an increased federal funds rate is that banks increase the rates that they charge their customers to borrow money.

Individuals are affected through increases to credit card and mortgage interest rates, especially if they carry a variable interest rate.

This has the effect of decreasing the amount of money consumers can spend.

This means that people will spend less discretionary money, which will affect businesses' top and bottom lines (that is, revenues and profits).

Therefore, businesses are also indirectly affected by an increase in the federal funds rate as a result of the actions of individual consumers.

But businesses are affected in a more direct way as well. They too borrow money from banks to run and expand their operations.

When the banks make borrowing more expensive, companies might not borrow as much and will pay higher rates of interest on their loans.

Less business spending can slow down the growth of a company, resulting in decreases in profit.

Clearly, changes in the federal funds rate affect the behavior of consumers and businesses, but the stock market is also affected.

Remember that one method of valuing a company is to take the sum of all the expected future cash flows from that company discounted back to the present.

To arrive at a stock's price, take the sum of the future discounted cash flow and divide it by the number of shares available.

This price fluctuates as a result of the different expectations that people have about the company at different times. Because of those differences, they are willing to buy or sell shares at different prices.

If a company is seen as cutting back on its growth spending or is making less profit - either through higher debt expenses or less revenue from consumers - then the estimated amount of future cash flows will drop.

All else being equal, this will lower the price of the company's stock. If enough companies experience declines in their stock prices, the whole market, or the indexes (that many people equate with the market, will go down.

[5]

c) Carry Trade:

Carry trade refers to borrowing in a low interest rate currency and investing in risk assets such as emerging market equities and debt. Indian markets, both equity and debt, have benefited from this over the past few years.

In the anticipation of the impending rate hike, the US dollar is likely to appreciate against other currencies—notably emerging market currencies including Indian currency.

This will damage carry trade, and liquidity in Indian capital markets.

A rate hike in the US will increase the cost of borrowing there and reduce the trade-off for this equation.

The differential between interest rates in the US and India is big, and a small hike in the US may not be attractive enough.

However, a signal from the US Fed will ultimately lead to subsequent rate hikes.

A series of hikes in interest rates in the US over a period of time will raise the borrowing cost for carry trade (borrow from US and invest in India), and thereby reduce their risk-adjusted return in India.

On the other hand, the Reserve Bank of India has embarked on cutting interest rates, and has cut repo rates thrice by 100 bps.

A cut in India and a hike in US would further reduce their risk-adjusted return. This may make US bonds more attractive.

The US is in any case considered a safe haven, and investors looking for stable returns will be more attracted towards US bonds.

[4]

[16 Marks]
